ANIRUDDHA DUTTA

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PROFESSIONAL SUMMARY

Lead Data Scientist and researcher combining production GenAI engineering with research in LLM safety, evaluation, and reasoning.

Specialized in Deep Learning, LLM architectures and evaluation, RAG pipelines, and agentic AI workflows.

Proven track record of translating advanced AI research into production-grade systems that drive measurable business impact and adoption at scale.

SKILLS

Generative AI & LLMs: GPT-4/5, LLaMA 3, Claude, Gemini, Qwen, DeepSeek, Prompt Engineering, Fine-tuning (LoRA/QLoRA), LLM Evaluation, LLM-as-Judge, Adversarial Testing, Prompt Injection Defense, Jailbreak Evaluation, Harm Detection Pipelines

RAG & NLP: LangChain, LangGraph, MCP Server, LlamaIndex, Vector DBs (FAISS, Pinecone, Solr), Embeddings, Semantic Search, Topic Modeling

Machine Learning & AI: XGBoost, Random Forest, LightGBM, Transformers, GNNs, LSTM, RNN, GRU

MLOps & Cloud: Databricks, MLflow, GCP Vertex AI, Snowflake CortexAI, CI/CD, Model Monitoring

Programming & Data: Python, PySpark, SQL, Pandas, NumPy, TensorFlow, PyTorch, HuggingFace

PROFESSIONAL EXPERIENCE

Lead Data Scientist | SimCorp, New York, NY — *Apr 2026 – Present*

- Leading the team's GenAI roadmap across data quality, safety evaluation, failure-mode detection, and system design — driving responsible AI deployment in regulated financial environments
- Built multi-agent AI systems, including agentic workflows for automated decision-making and debate-agent architectures to detect failure modes in multi-agent reasoning and power LLM-as-Judge evaluation

Senior Data Scientist | SimCorp, New York, NY — *Jan 2024 – Mar 2026*

- Built and deployed RAG pipelines with LangChain — custom PDF/XML parsing, semantic chunking, and vector embeddings — enabling natural-language querying across models and datasets. Improved retrieval recall and cut hallucinations by 10%.
- Built an LLM-as-Judge evaluation system with automated harm detection to catch hallucinations, retrieval failures, and policy violations in the RAG pipeline — cutting critical failures by 30% across 250+ enterprise clients.
- Built agentic AI workflows using LangChain, LangGraph and MCP servers, automating anomaly detection, decision-making, risk scoring and data processing pipelines improving data quality significantly
- Designed and optimized prompt engineering frameworks using CoT, few-shot learning, and self-consistency techniques; systematically tested 15+ prompt variations on financial text corpus, improving response accuracy by 15% across financial NLP use cases
- Fine-tuned LLMs (LLaMA 3, Qwen) using LoRA/QLoRA on 100K+ domain-specific datasets, increasing model precision and adoption across 250+ enterprise clients
- Led financial LLM benchmark research by developing an evaluation benchmark of 1,500 curated questions, implementing an LLM-as-judge framework with custom rubrics, and evaluating 10 LLM model families to address gaps in evaluation methodologies
- Built a multi-level classifier using Snowflake CortexAI to categorize private equity sectors and risk with 70% accuracy, developing FastAPI endpoints with Pydantic models and integrating into SimCorp One to process thousands of classifications monthly

Senior Quant Researcher | Axioma, New York, NY — *Apr 2021 – Dec 2023*

- Designed and implemented a real-time deep learning (LSTM/GRU) time series model with PyTorch, enhancing product performance by 5% and lowering model run times by 20%
- Engineered and deployed transformer-based time series models for portfolio optimization and backtesting, improving model KPIs by 5–10% across 1K+ scenarios/month; partnered with senior stakeholders across trading, risk, and product teams to define requirements and drive adoption across 250+ clients
- Engineered risk variables from unstructured text using advanced NLP techniques, which improved model KPI by ~5%

Artificial Intelligence Researcher | SumUp Analytics, San Francisco, CA *Oct 2019 – Dec 2019*

- Developed text clustering and summarization algorithms that improved NLP processing efficiency by 15% and cut output generation time by 30%
- Built topic modeling and sentiment analysis pipelines using NLTK across 100K+ financial and legal documents, surfacing social sentiment scores and business intelligence insights
- Implemented text classification and feature extraction pipelines that improved data labeling accuracy and downstream ML model performance

Data Scientist | Scotiabank, Toronto, Canada *Sep 2016 – Mar 2019*

- Developed and deployed credit risk models using XGBoost and Random Forest, improving prediction accuracy by 10% across 10K+ client records
- Built real-time fraud detection systems using LightGBM, enhancing detection accuracy and reducing financial risk exposure
- Designed and optimized graph neural networks (GNNs) for fraud detection, reducing false positives by 20%
- Performed feature engineering and model tuning, improving predictive model performance and stability
- Developed predictive analytics solutions for corporate bankruptcy risk assessment

Postdoc Data Scientist | Queen's University, Kingston, ON *Sep 2016 – Mar 2019*

Concurrent appointment with Scotiabank, above

EDUCATION

Ph.D., Computational Physics — University of Central Florida, Orlando, FL (2015)
M.S., Financial Engineering & Data Science — University of California, Berkeley (2020)
M.S., Physics — Indian Institute of Technology (IIT), Dhanbad, India (2008)

PUBLICATIONS & WORKING PAPERS

- Hasan, M., Dutta, A., Dutta, A. “When Agents Agree for the Wrong Reasons: Failure Modes in Multi-Agent Debate for Investment Reasoning” — in preparation for ACL 2027
  Introduces a taxonomy of collective failure modes — collusion, cascade, consensus suppression, epistemic capture — in multi-agent financial reasoning.
- Agarwal, Y., Dutta, A., Dutta, A. “FinTradeBench: A Financial Reasoning Benchmark for LLMs” — in review, NeurIPS 2026
  A benchmark for evaluating LLM financial reasoning, forming the foundation of a broader multi-agent AI safety research program.
- Dutta, A., Agarwal, Y., Dutta, A. “Diagnosing LLM-as-Judge Reliability for Document Relevance Assessment” — in preparation for NAACL 2027
  Finds LLM judges reach high inter-rater agreement while diverging from human ground truth on temporally sensitive financial documents.
- Dutta, A., Agarwal, Y., Dutta, A. “Does Stated Identity Bias LLM Investment Advice? A Situated-Interaction Audit of Recommendation Behavior” — in preparation for ACM FAccT 2027
  Audits whether a user's stated expertise level shifts LLM investment advice for identical underlying company fundamentals.
- Dutta, A., Batabyal, T., Acton, S.T. “An Efficient Convolutional Neural Network for Coronary Heart Disease Prediction” — Expert Systems with Applications, 2020
- Dutta, A., Kumar, S., Basu, M. “A Gated Recurrent Unit Approach to Bitcoin Price Prediction” — Journal of Financial Risk Management, 2020

Full list available on Google Scholar